Advanced Trigonometry

Compound angle formulas

1 Expand $\sin(x+30^\circ)$ in terms of $\sin x$ and $\cos x$, using exact values for $\sin 30^\circ$ and $\cos 30^\circ$.

2 Given $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, with A and B both acute, find $\cos(A+B)$.

3 Use the compound-angle formula for tangent to find the exact value of $\tan 75^\circ$, using $75^\circ = 45^\circ + 30^\circ$.

4 Use the compound-angle formula for cosine to find the exact value of $\cos 15^\circ$, using $15^\circ = 45^\circ - 30^\circ$.

Double angle formulas

5 Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\sin 2\theta$ and $\cos 2\theta$.

6 Given $\cos\theta = -\frac{1}{3}$ with $90^\circ < \theta < 180^\circ$, find $\cos 2\theta$ and state which quadrant 2θ lies in.

7 Given $\tan\theta = \frac{1}{2}$ with θ acute, find $\tan 2\theta$, using $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

Solving equations using double angle formulas

8 Solve $\cos 2\theta = \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

9 Solve $\sin 2\theta = \cos\theta$ for $0^\circ \le \theta < 360^\circ$, using $\sin 2\theta = 2\sin\theta\cos\theta$.

The R sin(θ+α) form

10 Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta+\alpha)$, with $R>0$ and $0^\circ < \alpha < 90^\circ$, giving α to the nearest degree.

11 Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta-\alpha)$, giving exact values of R and α.

Using the R-form to find extrema and solve equations

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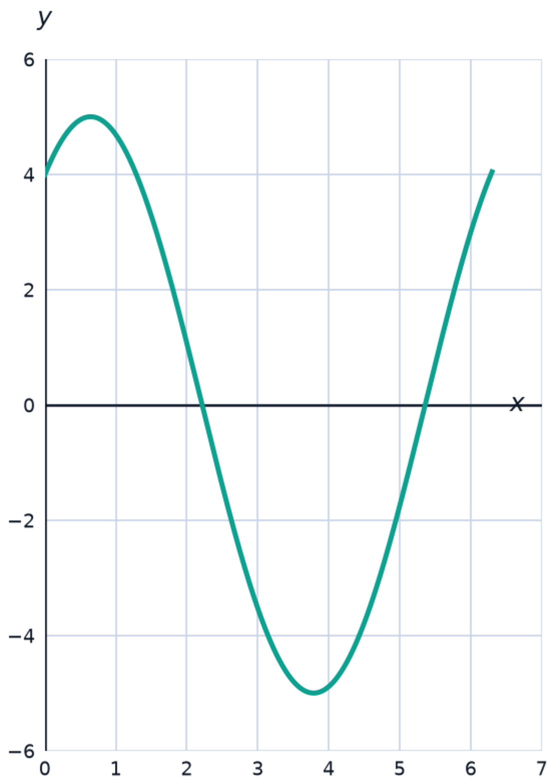
12 The graph shows $y = 5\sin(x + 53^\circ)$, the R-form of $3\sin x + 4\cos x$ found earlier. State the maximum and minimum values of $3\sin\theta + 4\cos\theta$, and the value of θ (to the nearest degree, $0^\circ \leq \theta < 360^\circ$) at which the maximum occurs.



13 Solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$, using the R-form $2\sin(\theta - 30^\circ)$ found earlier.

Proving trigonometric identities

14 Prove that $\dfrac{\sin 2\theta}{1 + \cos 2\theta} = \tan\theta$.

15 Prove that $\cos^4\theta - \sin^4\theta = \cos 2\theta$.

16 Prove that $(\sin\theta + \cos\theta)^2 = 1 + \sin 2\theta$.

Solving equations that combine several identities

17 Solve $2\cos 2\theta + 3\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$, giving answers to one decimal place where needed.

18 Solve $\cos 2\theta = 1 - \sin\theta$ for $0^\circ \leq \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

A well-designed word problem: alternating current

19 This question has three parts. The voltage in an AC circuit is modeled by $V(t) = 6\sin(100\pi t) + 8\cos(100\pi t)$ volts, where t is time in seconds. (a) Express $V(t)$ in the form $R\sin(100\pi t + \alpha)$, giving R exactly and α in radians to four decimal places. (b) State the maximum voltage reached, and the smallest positive value of t (in seconds, to four significant figures) at which it occurs. (c) Find the two values of t in $0 \le t < 0.02$ at which $V(t) = 5$, to four significant figures.

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